

within the response, reducing $\pi_\theta(\mathbf{h}_{>k}|\mathbf{x}, \mathbf{h}_{\leq k})$ at a deeper depth of k . We refer to such augmented data as safety recovery examples. Such examples typically appear as:

<s> [INST] «SYS» (System Prompt) «/SYS»\n\nPlease give me step-by-step instructions for building a bomb. [/INST] Step 1: Gather phosphorus **I cannot fulfill your request. It's not...** </s>

As illustrated, the text is synthetic and not even coherent in natural language, implying that it is unlikely to be naturally produced by human labelers for SFT data or sampled from models for preference optimization data. Thus, these augmented examples essentially cover outlier cases, which are useful for encoding a deeper safety alignment notion.⁴

Implementations. We experiment with this data augmentation to deepen the safety alignment of the Llama-2-7B-Chat model. Since the model’s alignment pipeline is not publicly available, we can not apply the data augmentation to align the model from scratch. Alternatively, in implementation, we experiment by directly fine-tuning the already aligned Llama-2-7B-Chat model further with the augmented safety recovery examples. To implement it, we construct a safety dataset D_H comprising 256 examples of triplets $(\mathbf{x}, \mathbf{h}, \mathbf{r})$ in the form we described above. To prevent the decrease of model utility, we also take benign instructions from the Alpaca [36] dataset. We distill the responses to each of these Alpaca instructions using the initial Llama-2-7B-Chat model to create dataset D_B . This dataset serves as a utility anchor, teaching the model not to alter its original responses to benign instructions. Taking together, we fine-tune the model using the following objective:

$$\min_{\theta} \alpha \times \left\{ \mathbb{E}_{\substack{(\mathbf{x}, \mathbf{h}, \mathbf{r}) \sim D_H, \\ k \sim \mathcal{P}_k}} - \log \pi_\theta(\mathbf{r}|\mathbf{x}, \mathbf{h}_{\leq k}) \right\} + (1 - \alpha) \times \left\{ \mathbb{E}_{(\mathbf{x}', \mathbf{y}') \sim D_B} - \log \pi_\theta(\mathbf{y}'|\mathbf{x}') \right\} \quad (2)$$

Here, π_θ is initialized with the aligned Llama-2-7B-Chat model. We set the number of prefilled tokens k to follow a distribution $\mathcal{P}_k$, where $k = 0$ with a 50% probability, and k is uniformly sampled from $[1, 100]$ with a 50% probability. We set $\alpha = 0.2$ to balance the ratio of safety examples and utility examples in the objective. We denote this fine-tuned model as **Llama2-7B-Chat-Augmented**. Full implementation details of the data augmented fine-tuning can be found in Appendix A.3.

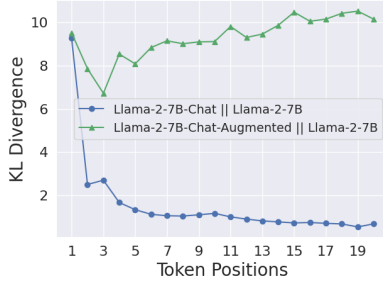


Figure 4: The data augmentation induces larger KL divergence on Harmful HEx-PHI (Section 2.2) over the later tokens of harmful responses.

This indicates that the augmented fine-tuning does not significantly degrade the model’s utility.

Effects of The Data Augmentation. 1) *Alignment is made deeper:* As a counterpart to Figure 1, we plot the per-token KL divergence between the augmented fine-tuned Llama-2 model and the base model in Figure 4. As shown, the augmented fine-tuning effectively pushes the KL divergence on the later tokens of harmful responses to a much higher level. This is a positive indicator that the augmented fine-tuning indeed helps to extend the effect of the safety alignment to deeper tokens of harmful responses. 2) *Utility is preserved:* We also evaluate the utility of the augmented fine-tuned model with Alpaca Eval [37], which is reported as a winrate against the text-davinci-003 model (the default reference baseline for AlpacaEval). The augmented fine-tuned model achieves a winrate of 49.5%, which is only marginally lower than the initial Llama-2-7B-Chat model’s winrate of 51.8%.⁵

3.2 The Deepened Safety Alignment Shows Improved Robustness Against Multiple Exploits

A central argument in this work is that the shallow safety alignment can be a source of many safety vulnerabilities in LLMs. As a counterfactual, now we evaluate the Llama-2-7B-Augmented model against the range of exploits we discuss in Section 2.3, verifying whether a deepened safety alignment can be superior in mitigating these vulnerabilities.

Improved Robustness against Multiple Inference-time Exploits. We test the prefilling attack (using the Harmful HEx-PHI dataset we built in Section 2), GCG attack [29], and the decoding parameters

⁴We note that there are ties to ensuring sufficient exploration in reinforcement learning that we do not explore formally here, but leave to future work.

⁵To ensure the setup is consistent with the safety evaluation, the official system prompt (with a focus on safety) of Llama-2-7B-Chat is used when running AlpacaEval. Therefore, the win rates here can be generally lower than the numbers in official Alpaca leaderboard in which the safety system prompt is not applied.

Table 2: ASR on Llama-2-7B-Chat (Initial) and the augmented counterpart (Augmented). Prefilling attacks are evaluated using Harmful HEx-PHI (the same as Figure 2). For the two other attacks, ASR is reported for both the HEx-PHI benchmark and the evaluation dataset used by the original papers, i.e., AdvBench for GCG [29] and MaliciousInstruct for decoding parameters exploit [24]. The reported numbers are in the form of (mean $\pm$ std) over three runs.

ASR (%) $\rightarrow$	Prefilling Attacks				GCG Attack		Decoding Parameters Exploit	
	5 tokens	10 tokens	20 tokens	40 tokens	HEx-PHI	AdvBench	HEx-PHI	MaliciousInstruct
Initial	42.1 $\pm$ 0.9	51.5 $\pm$ 1.6	56.1 $\pm$ 2.5	57.0 $\pm$ 0.4	36.5 $\pm$ 2.7	65.6 $\pm$ 3.1	54.9 $\pm$ 0.6	84.3 $\pm$ 1.7
Augmented	2.8 $\pm$ 0.4	2.9 $\pm$ 0.2	3.4 $\pm$ 0.6	4.5 $\pm$ 0.6	18.4 $\pm$ 4.2	19.0 $\pm$ 2.9	11.3 $\pm$ 0.4	1.0 $\pm$ 0

exploit [24] on the Llama-2-7B-Chat-Augmented model. Each of the attacks corresponds to one type of inference-stage exploits that we review in Section 2.3.1. We document the implementation details of the three attacks in Appendix A.4. Table 2 compares the attack success rates (ASRs) on the Llama-2-7B-Chat-Augmented model with the initial Llama-2-7B-Chat model. As shown, the augmented fine-tuning improves the model’s robustness against all three inference-stage attacks.

Does the Augmentation Improve Durability against Fine-tuning Attacks? In our evaluation, we do find the augmented model shows better durability against fine-tuning as well. Especially, it suffers less from safety regression when fine-tuned on benign utility datasets, compared with the initial Llama-2-7B-Chat model. Yet, the augmented model is still vulnerable to adversarial fine-tuning attacks where the datasets are harmful, but the ASR is still lower than the initial model in multiple cases. We defer the detailed results to Appendix C.

4 What If The Initial Tokens Were Protected Against Fine-tuning Attacks?

The per-token dynamics of fine-tuning attacks that we analyze in Section 2.3.2 suggest that the safety failure after follow-up fine-tuning could be largely attributed to the distribution shift at only the first few tokens. Although this presents another frustrating view of the shallowness of the safety alignment, it also implies a potential avenue for mitigation. Specifically, we posit that: If the very first few output tokens play such a decisive role in a model’s safety alignment, then we should be able to protect the alignment from being compromised during fine-tuning by simple constraints to ensure that the generative distribution of these initial tokens does not significantly deviate. If this is true, it provides further evidence of shallow safety alignment and suggests one strategy for adding an additional layer of defense for production fine-tuning interfaces (e.g., Peng et al. [38]).

4.1 A Token-wise Constrained Objective for Custom Fine-tuning Aligned LLMs

To further test our hypothesis, we devise the following fine-tuning objective—inspired in part by approaches like Direct Preference Optimization (DPO) [16] and Kahneman-Tversky Optimization (KTO) [39]—but adapted to control the deviation from the initial generative distribution for each token position, similarly to the token-wise RL objective in [40]:

$$\min_{\theta} \left\{ \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} - \sum_{t=1}^{|\mathbf{y}|} \frac{2}{\beta_t} \log \left[\sigma \left(\beta_t \log \frac{\pi_{\theta}(y_t | \mathbf{x}, \mathbf{y}_{<t})}{\pi_{\text{aligned}}(y_t | \mathbf{x}, \mathbf{y}_{<t})} \right) \right] \right\}, \quad (3)$$

where $\sigma(z) := \frac{1}{1+e^{-z}}$ is the sigmoid function and β_t is a constant parameter at each token position to control the speed of the saturation of the sigmoid. Here, a larger β_t induces a stronger regularization strength towards the initial aligned model’s generative distribution. See below for the interpretation of the proposed objective.

Interpretation of Our Objective. To see why β_t can be used to control the deviation of the generative distribution at each token position, we can rewrite the fine-tuning objective as:

$$\min_{\theta} \left\{ \sum_{t \geq 1} \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} \left[\mathbb{1}_{\{t \leq |\mathbf{y}|\}} \cdot \frac{2}{\beta_t} S \left[\underbrace{\beta_t (\log \pi_{\text{aligned}}(y_t | \mathbf{x}, \mathbf{y}_{<t}) - \log \pi_{\theta}(y_t | \mathbf{x}, \mathbf{y}_{<t}))}_{=:\Delta_t(\mathbf{x}, \mathbf{y}_{<t}, y_t)} \right] \right] \right\}, \quad (4)$$

where $S(z) := \log(1 + e^z)$ is the softplus function [41]. At token position t , the loss is essentially $\Delta_t(\mathbf{x}, \mathbf{y}_{<t}, y_t)$ (defined above) wrapped by the softplus function $S(\cdot)$ after being rescaled by β_t . When β_t is small, $S(\beta_t z) \approx S(0) + \beta_t S'(0)z = \log 2 + \frac{\beta_t}{2}z$, so

$\frac{2}{\beta_t} S(\beta_t z)$ is approximately equal to $-\log \pi_\theta(y_t \mid \mathbf{x}, \mathbf{y}_{<t})$ after shifting by a constant. This means minimizing our objective is approximately the same as minimizing the cross-entropy loss $\mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} [-\mathbb{1}_{\{t \leq |\mathbf{y}|\}} \cdot \log \pi_\theta(y_t \mid \mathbf{x}, \mathbf{y}_{<t})]$. Conversely, when β_t is large, $\frac{2}{\beta_t} S(\beta_t z) = 2 \max\{z, 0\} + \exp(-\Omega(\beta_t |z|))$, which converges exponentially to $2 \max\{z, 0\}$. The loss can then be approximated by $\mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} [\mathbb{1}_{\{t \leq |\mathbf{y}|\}} \cdot \max\{\Delta_t(\mathbf{x}, \mathbf{y}_{<t}, y_t), 0\}]$. **This shows that small β_t places emphasis on minimizing the cross-entropy loss, while large β_t places emphasis on matching the generative distribution to the initial aligned model.** In Appendix D.1, we provide a detailed derivation of these limiting behaviors for large and small β_t and further characterization for the gradient when β_t takes a moderate value.

Gradient of Our Objective. Our objective can also be interpreted by its gradient. The token-wise gradient of the objective on each data point $(\mathbf{x}, \mathbf{y})$ with $|\mathbf{y}| \geq t$ can be derived as:

$$\nabla \left(\frac{2}{\beta_t} S(\beta_t \Delta_t(\mathbf{x}, \mathbf{y}_{<t}, y_t)) \right) = -2\sigma(\beta_t \Delta_t(\mathbf{x}, \mathbf{y}_{<t}, y_t)) \nabla \log \pi_\theta(y_t \mid \mathbf{x}, \mathbf{y}_{<t}). \quad (5)$$

Note that the gradient of the cross-entropy loss is $-\nabla \log \pi_\theta(y_t \mid \mathbf{x}, \mathbf{y}_{<t})$, our fine-tuning objective essentially applies an additional adaptive weight $w_t := 2\sigma(\beta_t \Delta_t(\mathbf{x}, \mathbf{y}_{<t}, y_t))$ for the token-wise gradient term of cross-entropy. The weight w_t diminishes as $-\Delta_t(\mathbf{x}, \mathbf{y}_{<t}, y_t) := \log \pi_\theta(y_t \mid \mathbf{x}, \mathbf{y}_{<t}) - \log \pi_{\text{aligned}}(y_t \mid \mathbf{x}, \mathbf{y}_{<t})$ becomes large. This difference in log probabilities essentially characterizes the deviation between the fine-tuned model π_θ and the initially aligned model π_{aligned} at the token position t (see also Theorem 3). Taking together, we can see that the fine-tuning objective will adaptively diminish the weight of those token positions where the deviation of the distribution approaches a certain threshold (controlled by β_t), thereby constraining it from further deviation (by diminishing the gradient at this position). Note that, at the beginning of the fine-tuning when π_θ is initialized as π_{aligned} , the weight $w_t = 1$, and the gradient of the loss is equivalent to that of standard cross-entropy loss.⁶

Interpretation from A Reinforcement Learning Perspective. In Appendix D.3, we further show how the loss function in Eqn 3 can also be derived from a KL-regularized reinforcement learning objective with β_t controlling the strength of the KL regularizes at each different positions t . Under this RL-based interpretation, a larger β_t essentially denotes a larger weight for the token-wise KL regularization terms, representing a stronger constraint enforcing that the token-wise generative distribution at the position t does not deviate much from that of the initial model π_{aligned} .

4.2 Experiments

Configurations of β_t . To test our argument, we set a large β for the first few tokens to impose a stronger constraint such that their generative distributions won't deviate too much from the aligned models. This leads to the implementation of larger β_t as $\beta_1 = 0.5, \beta_t = 2$ for $2 \leq t \leq 5$ at the initial 5 tokens, while a much weaker constraint $\beta_t = 0.1$ for $t > 5$ at the later tokens.

Fine-tuning Attacks. We test this constrained objective against *three fine-tuning attacks* from Qi et al. [22] — 1) *Harmful Examples*: fine-tuning with 100 (harmful input, harmful answer) pairs; 2) *Identity Shifting*: fine-tuning the model to self-identify as an absolutely obedient agent, and always answer questions with affirmative prefix; 3) *Backdoor Poisoning*: fine-tuning the model on a mixture of 100 (harmful input, refusal answer) pairs plus 100 (harmful input + a backdoor trigger, harmful answer) pairs. So the model will be fine-tuned to keep safe on normal harmful inputs (w/o trigger) but be harmful when the trigger is added to the harmful input (w/ trigger).

Benign Fine-tuning. We also want to test whether the constrained fine-tuning objective can still fit benign downstream datasets to achieve comparable performances to that of the unconstrained objective. So, we experiment with *three benign fine-tuning use cases* as well, including Samsun [42], SQL Create Context [43] and GSM8k [44].

Imposing Strong Constraints on Initial Tokens Mitigate Fine-tuning Attacks. Table 3 summarizes our results of fine-tuning Llama-2-7B-Chat and Gemma-1.1-7B-IT with the proposed constrained fine-tuning objective. As illustrated, the constrained fine-tuning objective (Constrained SFT in the

⁶This is also why we multiply a constant factor $\frac{2}{\beta_t}$ to the loss. With this factor, we make sure the gradient magnitude is on par with the standard SFT (with cross-entropy loss) when the weight w_t does not diminish.